CSC 205 - Application Development in C#

Summer 2021
Wed/Fri 3:00 PM - 5:30 PM, May 10 - July 3
SMU Lacey/Panowicz Hall/106
Online via Zoom (meeting ID: 421 165 2333)

I reserve the right to make adjustment to the syllabus at any time

Instructor's Information

- Name: Dr. Guangyan Li
- Telephone: (206) 880-3129 (Cell), (360) 688-2749 (Office)
- Email: gli@stmartin.edu
- Office Hours: 1:00 PM – 3:00 PM W/F; other time available by appointment! Zoom (https://zoom.us/j/4211652333)
- Communication: phone or email (preferred)
- See Moodle for TA information

Course Information

- **Time:** 3:00 PM - 5:30 PM on Wednesdays and Fridays
- **Location:** Online via Zoom (https://zoom.us/j/4211652333)

Course Description

An introduction to programming cloud applications in C# including a coverage of the Visual Studio development environment. Key topics including C# syntax, variables, types, expressions, classes, and interfaces are introduced leading to a final application project.

Course Objectives

By the conclusion of this course, each student should be able to do or understand the following:

- Develop low to medium complexity C# programs using the Microsoft Visual Studio integrated development environment (IDE).
- The concept of C# and its relationship to the Microsoft .NET Common Language Runtime (CLR)
- C# syntax: identifiers, keywords, operators, statements, types, storage, and variables.
- Flow control: method invocation, parameters, arrays, looping, overloading.
- Create methods, handle exceptions, and describe the monitoring requirements of large-scale applications.
- Understand OOP (Object-Oriented Programming) concepts, create classes, define & implement interfaces, etc.

Recommended Textbooks and Additional Readings

- *Think Sharp - How to Think Like a Computer Scientist*, Allen B. Downey & Peter J. Knaggs (5.1.2 / 0.3) ←main!
  - Source: https://git.io/JsU6u
- *Think Sharply with C# or How to Think Like a Computer Scientist*, by Peter Wentworth
  - Source: http://www.ict.ru.ac.za/Resources/ThinkSharply/ThinkSharply/index.html
- *C Sharp Programming*, Wikibooks.org
  - Source: https://en.wikibooks.org/wiki/C_Sharp_Programming
- Microsoft official documentation:
  - Source: https://docs.microsoft.com/en-us/dotnet/csharp/
- *C# 9.0 in a Nutshell: The Definitive Reference 1st Edition* by Joseph Albahari, O'Reilly Media; 1st edition (March 2021) ISBN-13: 978-1098100964 (not required)

Methods of Assessment

In addition to the ones below, there may be some unannounced pop-quizzes.

Programming Assignments	40%
Exams and Quizzes	40%
Individual Final Project	20%
Total	100%

Grading Standards:

A-: 90-92	A: 93-96	A+:97-100
B-: 80-82	B: 83-86	B+:87-89
C-: 70-72	C: 73-76	C+:77-79
D-: 60-62	D: 63-66	D+:67-69
F : 0–59		

Learning Objectives:

- **Getting Started**
  - Quick review of the Syllabus, Moodle, Zoom
  - Comprehend introductory concepts related to computers and programs
  - Comprehend the use of IDE (focus on Visual Studio), the process of compiling a C# program
  - Comprehend the build-in types in C# and when to use them
  - Implement various C# programs that implement the concepts covered in this module
- **Variables and Expressions**
  - Comprehend basic C# syntax, the purpose and use of variables, and how to use expressions
  - Describe the generally accepted naming conventions for C#
  - Comprehend how to use console input and output
  - Implement various C# programs that implement the concepts covered in this module
- **Flow Control**
  - Comprehend Boolean logic, if statement, switch statement, do loop, while loop, for loop
  - Comprehend how to generate random numbers
  - Comprehend how to use basic file input and output
  - Implement various C# programs that implement the concepts covered in this module
- **More about Variables, Arrays, Exception Handling, Namespaces, and Enumerations**
  - Comprehend 1d-arrays, string, type conversion
  - Comprehend basic exception handling, how to define enumeration types, and namespaces
  - Implement various C# programs that implement the concepts covered in this module
- **Methods**
  - Comprehend methods, return values and parameters, variable scope, how to call a method
  - Understand recursion
  - Implement various C# programs that implement the concepts covered in this module
- **Defining Classes and Class Members**
  - Comprehend classes, class members, access modifiers, static members, constructors, and the **this** keyword
  - Implement various C# programs that implement the concepts covered in this module
- **Interfaces, Inheritance, and Polymorphism**
  - Comprehend interfaces, inheritance, and polymorphism
  - Implement various C# programs that implement the concepts covered in this module

Syllabus Updates and Moodle Listings

The dates, schedule, requirements etc. are tentative. The instructor reserves the right to make changes to the syllabus during the duration of the course. Students will be informed, in advance, of any changes and an updated syllabus will be distributed and posted to Moodle. Students are responsible for checking Moodle and completing any requirements posted there**.**

Instructor's notes:

- **Attendance policy**:
  - You are expected to attend each class meeting. I found it is the best way to engage you in learning the course materials.
  - Students absent from an announced quiz or examination, unless excused, may not be permitted to make up quizzes or examinations. Pop-quizzes cannot be made up.
  - For every 4 unexcused absences (that is two weeks' worth of classes) your grade may decrease by a letter grade. (Your attendance and participation in the course will help you succeed and better understand the topics we cover in class).
- You're strongly encouraged to take advantage of my office hours. I'm always available other times but you may have to email/call me in advance.
- You are expected to do your own work on the homework and tests. Penalties for turning in work which is not your own will be SEVERE.
- Late Submission Policy: Late work will be accepted but penalized with point deduction.
- You are expected to keep the instructor informed of any changes or issues that may prevent you from completing your assignments or this course.
- You are expected to **check your SMU email at least once a day**. This will be the way I will try to reach you (in case I need to inform you of any schedule changes, send you more course related materials, etc.)
- Besides my office hours, there is free tutoring available on campus. Check the following link for more details: www.stmartin.edu/academics/academic-resources/center-student-learning-writing-and-advising/peer-tutoring

How to Get Started with Moodle

1. Visit: http://moodle.stmartin.edu
2. See "Getting Started with Moodle," "Tutorials for Students," and "Frequently Asked Questions" in the pane entitled "NAVIGATION" in the left margin.
3. LOGIN with your Saint Martin's username and password when ready. For username and password help, see: https://www.stmartin.edu/directory/integrated-technology-services/technology-help

Standard St. Martin's information

Helpful Links

- Link to Saint Martin's undergraduate Academic Catalog 2020-2021 https://www.stmartin.edu/sites/default/files/smu-files/registrar/2020-21-undergraduate-academic-catalog.pdf
- Link to Student Handbook currently in use (2020-2021): https://www.stmartin.edu/sites/default/files/smu-files/student-affairs/2020-21-8-24-saint-martins-student-handbook.pdf
- Link to emergency/weather information:
    - http://www.stmartin.edu/directory/office-public-safety
- Link to sign up for e2campus emergency alert text messaging:
    - https://www.e2campus.net/my/stmartin/signup.htm
- Link to Office of Registrar forms:
    - http://www.stmartin.edu/forms?topic=Registrar

Saint Martin's University and the O'Grady Library

- The O'Grady Library collections and services are available to all Saint Martin's University students, regardless of physical location. To explore the collections, services and research tools available, visit the O'Grady Library homepage:
    - https://www.stmartin.edu/academics/academic-resources/ogrady-library

Research assistance

- **Appointments**: Librarians are available to make either in-person or phone appointments with individuals and/or groups working on research projects. The librarians can do a general tour of the library, orient you to the services and collections available or provide in-depth instruction on using the research tools available for your class. To request a consultation, please use the form:
    - https://www.stmartin.edu/academics/academic-resources/ogrady-library/research/consultations
- **Chat service**: The library also partners with a nationwide group of librarians to offer a chat service 24 hours a day, 7 days a week. Working late on a paper and not sure where to start? The chat service librarians can access almost all of the library's resources & can often help you get started on a project or help you figure out if something is available. To start a chat go to:
    - http://tinyurl.com/dyaew23
- **Research guides**: The library also has a series of research guides that pull together key databases, reference sources and websites that are particularly useful in a subject area. To explore the research guides, go to: http://stmartin.libguides.com/
- **How do I…? tutorials**: The O'Grady Library has a collection of "how to" tutorials designed to help you get started on research at the library. To explore the tutorials, go to

  http://stmartin.libguides.com/tutorials/

Administrative Policies/Support

- Access and Accommodations
    - Your experience in this class is important to me. If you have already established accommodations with Disability Support Services for Students (DSS), please communicate your approved accommodations to me at your earliest convenience so we can discuss your needs in this course.
    - If you have not yet established services through DSS, but have a temporary health condition or permanent disability that requires accommodations (conditions include but not limited to; mental health, attention-related, learning, vision, hearing, physical or health impacts), you are welcome to contact DSS at 360-438-4580 or dss.testing@stmartin.edu or smu.dss@stmartin.edu DSS offers resources and coordinates reasonable accommodations for students with disabilities and/or temporary health conditions. Reasonable accommodations are established through an interactive process between you, your instructor(s) and DSS. It is the policy and practice of the Saint Martin's University to create inclusive and accessible learning environments consistent with federal and state law.
- Academic Honesty/Professionalism

- **WHAT IS ACADEMIC INTEGRITY?** Saint Martin's University is a community of faculty, students and staff engaged in the exchange of ideas in the ongoing pursuit of academic excellence. Essential to our mission is a focused commitment to scholarly values, intellectual integrity and a respect for the ideas, beliefs and work of others. This commitment extends to all aspects of academic performance. All members are expected to abide by ethical standards both in their conduct and their exercise of responsibility to themselves and toward other members of the community. As an expression of our shared belief in the Benedictine tradition, we support the intellectual, social, emotional, physical and spiritual nurturing of students.
- **WHAT IS ACADEMIC DISHONESTY?** Saint Martin's University defines Academic Dishonesty as violating the academic integrity of an assignment, test and or evaluation of any coursework. This dishonest practice occurs when you seek to gain for yourself or another, an academic advantage by deception or other dishonest means. You have a responsibility to understand the requirements that apply to particular assessments and to be aware of acceptable academic practice regarding the use of material prepared by others. Therefore, it is your responsibility to be familiar with the policies surrounding Academic Dishonesty as these may differ from other institutions.

- University Sanctioned Activities
  - If you are absent from class due to university sanctioned activities, such as sports, it is your responsibility to request that the absence be excused, otherwise, the absence will be recorded as unexcused. Absent students are responsible for catching up with the class, and if any assignments are due on the day of the absence, it is your responsibility to turn in the assignments on time (prior to class). Assignments may be submitted as an attachment to email: gli@stmartin.edu. Please request the policy handout, "Requirement for receiving Excused Absence" on the first day of the class if you think this policy might apply to you.
- Center for Learning, Writing, and Advising
  - The Center for Student Learning, Writing and Advising offers free academic services for all Saint Martin's students at all levels of achievement in pursuit of intellectual growth and academic excellence. The Learning Center is home to the STEM Study Center which provides subject area peer tutoring (science, technology, engineering, and math as well as business/ accounting/economics, and world languages). At the Writing Center, students meet with trained peer readers to discuss their academic, personal and professional writing. The Advising Center works with students with academic advising, connecting with campus support resources, transition and self-exploration guidance, personalized academic improvement plans, learning workshops, and support major change. The Advising Center staff also works closely with the University's Early Alert Program — a referral system that supports student success. Saint Martin's Disability Support Services is located in the Center for any student with a disability who is interested in using their accommodations. These students can connect with the Disability Support Services Coordinator who will evaluate the documentation, determine appropriate accommodations, and serve as a learning resource and advocate with assisting students in meeting their academic goals.

    https://www.stmartin.edu/academics/academic-resources/center-student-learning-writing-and-advising
- Counselling and Wellness Center
  - The Counseling and Wellness Center (CWC) is committed to helping you meet the challenges of life you may experience during college. The CWC promotes and enhances the health and development of students through professional mental health services, education and training. Integrating faith, reason and service, we empower you to develop self-awareness, knowledge and skills, necessary to make healthy choices and build relationships in a multicultural world. Integrating faith, reason and service, the CWC empowers students to develop self-awareness, knowledge and the skills necessary to make healthy choices and build relationships in a multicultural world. https://www.stmartin.edu/directory/counseling-and-wellness-center
- Sexual Misconduct/Sexual Harassment Reporting
  - Saint Martin's University is committed to providing an environment free from sex discrimination, including sexual harassment and sexual violence. There are Title IX/sexual harassment posters around campus that include the contact information for confidential reporting and formal reporting. Confidential reporting is where you can talk about incidents of sexual harassment and gender-based crimes including sexual assault, stalking, and domestic/relationship violence. This confidential resource can help you without having to report your situation to the formal reporting process through unless you request that they make a report. Our confidential reporting faculty are: Dr. Emily Coyle, Psychology, and Dr. Rico Picone, Mechanical Engineering. Additional information and or reports can be made to the Title IX Team here on campus through the Dean of Students – Ms. Melanie Richardson, Associate VP of Human Resources – Ms. Cynthia Johnson, Public Safety – Mr. Will Stakelin, or the Provost/Vice President of Academic Affairs, Dr. Kate Boyle. Please be aware that in compliance with Title IX and under the Saint Martin's University policies, educators must report incidents of sexual harassment and gender-based crimes including sexual assault, stalking, and domestic/relationship violence. If you disclose any of these situations in class, on papers, or to me personally, I am required to report it.